WEILI XU

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Research Overview

I am a senior undergraduate in Computer Engineering in the UIUC–Zhejiang University dual-degree program. **I am seeking Ph.D. opportunities in machine learning systems (MLSys) starting in Fall 2027.**

My research focuses on agentic LLM serving, hardware-aware sequence modeling, and efficient long-context multimodal systems. My recent work includes **Evolving Infra** (co-first author), on agent-adaptive serving, and **FlashStates** (co-first author and project lead), on hybrid state management.

Education

B.S. — **University of Illinois Urbana-Champaign**
Aug. 2023 – Present — Computer Engineering; GPA: 3.97/4.00

B.S. — **Zhejiang University**
Aug. 2023 – Present — Computer Engineering (Dual Degree); GPA: 4.20/4.30

Selected coursework (all A/A+): ECE 408 Applied Parallel Programming; ECE 391 Computer Systems Engineering.

Employment and Research

Research Intern — **Together AI**
May – Sep. 2026 — Frontier Performance Team, San Francisco, CA
QATFactory (co-first author): deployment-aligned quantization-aware training and distillation of LLMs.

Research Intern — **University of Illinois Urbana-Champaign**
Aug. 2025 – Present — Machine Learning Systems, Urbana, IL
ThunderAgent: program-aware scheduling to address KV-cache thrashing and cross-node memory imbalance. Co-first author; ICML 2026 Spotlight. Achieved 1.5–3.6× serving and 1.8–3.9× RL rollout throughput on 8×H100.

Research Intern — **Zhejiang University**
Aug. 2023 – Jul. 2025 — Machine Learning, Hangzhou, China
AuroraLong: first-author work on efficient hour-long video understanding (ICCV 2025), with 8.13× faster inference on RTX 3090. Video-MMLU: benchmark evaluating 90+ video LLMs; ICCV workshop Outstanding Paper Award and Oral Presentation.

Teaching

Teaching Assistant — **ZJU–UIUC Institute**
Aug. 2024 – May 2025 — ECE 220 Computer Systems & Programming (Spring 2025)
ECE 120 Introduction to Computing (Fall 2024)

Selected Publications

The * sign denotes equal contribution. The † sign denotes project lead. Paper titles link to the corresponding publications.

Peer-Reviewed Papers

C2 Hao Kang*, Ziyang Li*, Weili Xu*, Xinyu Yang*, Yinfang Chen, Junxiong Wang, Beidi Chen, Tushar Krishna, Chenfeng Xu, and Simran Arora. ThunderAgent: A Simple, Fast and Program-Aware Agentic Inference System. *International Conference on Machine Learning (ICML)*, ***Spotlight***, 2026.

C1 Weili Xu, Enxin Song, Wenhao Chai, Xuexiang Wen, Tian Ye, and Gaoang Wang. Bringing RNNs Back to Efficient Open-Ended Video Understanding. *International Conference on Computer Vision (ICCV)*, 2025.

Workshop Papers

W1 Enxin Song, Wenhao Chai, Weili Xu, Jianwen Xie, Yuxuan Liu, and Gaoang Wang. Video-MMLU: A Massive Multi-Discipline Lecture Understanding Benchmark. *International Conference on Computer Vision (ICCV) Workshops*, 2025.

Preprints and Technical Reports

P3 Hao Kang*, Weili Xu*, Ziyang Li*, Zheng Du*, Hanchen Li, Tian Xia, Song Bian, Tushar Krishna, Huizi Mao, Ion Stoica, and Song Han. Evolving Infra: Agent-Adaptive LLM Serving at Scale. *Technical report, arXiv:2611.12304*, 2026.

P2 Chenxi Li*, Weili Xu*†, Tianyi Zhang, Jue Wang, Zhongzhu Zhou, and Chenfeng Xu. FlashStates: Unlocking Flexible Hybrid State Management for Agentic LLM Inference. *Preprint*, 2026.

P1 Weili Xu*, Jisen Li*, Yuqing Jian*, Chenxi Li, Zhizhou Sha, Yifan Yu, Qingyang Wu, Chenfeng Xu, Zhongzhu Zhou, Tianyi Zhang, and Ben Athiwaratkun. QATFactory: A Versatile, Deployment-Aligned Framework for Quantization-aware Training and Distillation of LLMs. *Preprint, arXiv:2609.39223*, 2026.

Open-Source Contributions

Contributor Dec. 2025 – Jun. 2026	**TileGym (NVIDIA's cuTile Kernel Library)** Implemented backward-pass FlashAttention and SwiGLU kernels in cuTile, with correctness tests and benchmarking; achieved >10% higher throughput than the optimized Triton baseline.

Technical Skills

Programming: Python, C/C++, Triton, CUDA, cuTile, Java, RISC-V Assembly.

Frameworks and systems: PyTorch, vLLM, SGLang, TensorRT-LLM, Dynamo, NIXL, Mooncake, slime, Megatron-LM.

Tools: Git, Linux/UNIX, Slurm, Kubernetes, Docker, GDB, PDB, Nsight Systems, Nsight Compute.